

ORCHESTRA-OS RESEARCH PROGRAM

ORCHESTRA-OS Testing and Release-Readiness Report

Benchmark evidence, improvement priorities, and release gates

Evidence boundary: current results are userspace-validated only. This PDF does not establish a kernel scheduler, causal performance advantage, production security, or deployment readiness.

Paper CPU userspace smoke v1 result - 2026-08-03

Evidence status

This is a bounded **userspace-validated smoke result** for selected WP2-WP6 mechanics. It is not an Experimentally validated scheduler result, a kernel prototype, a causal baseline comparison, a hard-real-time result, or production-security evidence.

The authoritative retained copy for this result is:

```
artifacts/test-results/2026-08-03/benchmark/authoritative-c
```

It is a byte-identical copy of the original `/tmp/orchestra-paper-cpu-smoke-v1-20260803-c` capture and contains all six raw CSV files and stderr logs, per-run records, provenance, validation outcomes, and processed summaries. The dated artifact index and `SHA256SUMS` preserve its local identity. Approved long-term storage and retention policy are still required for release evidence. The earlier `-a` and `-b` campaigns are retained separately as superseded evidence and must not be mixed with this result.

Frozen inputs and host

- Source SHA-256: `d5ec68c28a2cfcc20475dba44cb4b3c36b46b08acb4206647b874d2e2edf8713`
- Binary SHA-256: `f2f14020ef6f4caf7cc1f9dda3b392451367209b9d9f79e27415680c5e678b4b`
- Harness SHA-256: `3c07263066118a7f08c9e10ae80c488bb8fae680db08b31cba6b7154aec193d6`
- Manifest SHA-256: `40844535648f4a7dc25131141035d9cfb88886c20f0f8b9fb4bbbc8ffef02b5a`
- Metrics schema SHA-256: `04cb97e55d66bcf3e03e810372865c25368c89abb5de4e09e43d952d6f2d09c2`
- Captured at: `2026-08-02T19:33:41Z` (2026-08-03 Asia/Kolkata)
- Host: Intel Core i5-10310U, 4 cores / 8 threads, approximately 32 GB RAM
- Kernel: Kali Linux `7.0.12+kali-amd64`, x86-64, bare metal
- Frequency policy: `powersave`; Intel turbo was enabled
- Privilege: non-root, no effective capabilities; real-time entry was disabled
- Repository state: not a Git worktree, so file hashes provide revision identity

The binary was rebuilt immediately before the campaign with the repository Makefile (`cc -O2 -std=c11 -Wall -Wextra -Wpedantic -Werror ...`). The harness records source and binary hashes but cannot independently prove their build association; the command and matching pre/post hashes are the available local provenance because this directory has no Git metadata.

The host was an active desktop rather than an isolated benchmark machine. Background activity and the powersave policy are recorded in provenance and limit interpretation.

Protocol completion

The manifest ran four eligible workers, no real-time workers, one second of calibration, four measured seconds, a 100 ms period, ten excluded warm-up ticks, and three fixed seeds per mode. Baseline ran before ORCHESTRA for each seed; modes never ran concurrently.

All six invocations:

- returned zero without timeout or interruption;
- produced exactly 40 rows, with 30 post-warm-up rows;
- exceeded the predeclared 4.5-second minimum wall duration;
- passed the strict 40-column schema and cross-field validation;
- preserved source and binary hashes throughout the campaign;
- reported zero publisher deadline misses and zero frame rejections; and
- produced two bounded controller updates per ORCHESTRA invocation.

The statistical unit is one invocation, so each mode has `n = 3`; tick rows were not treated as independent observations.

Descriptive results

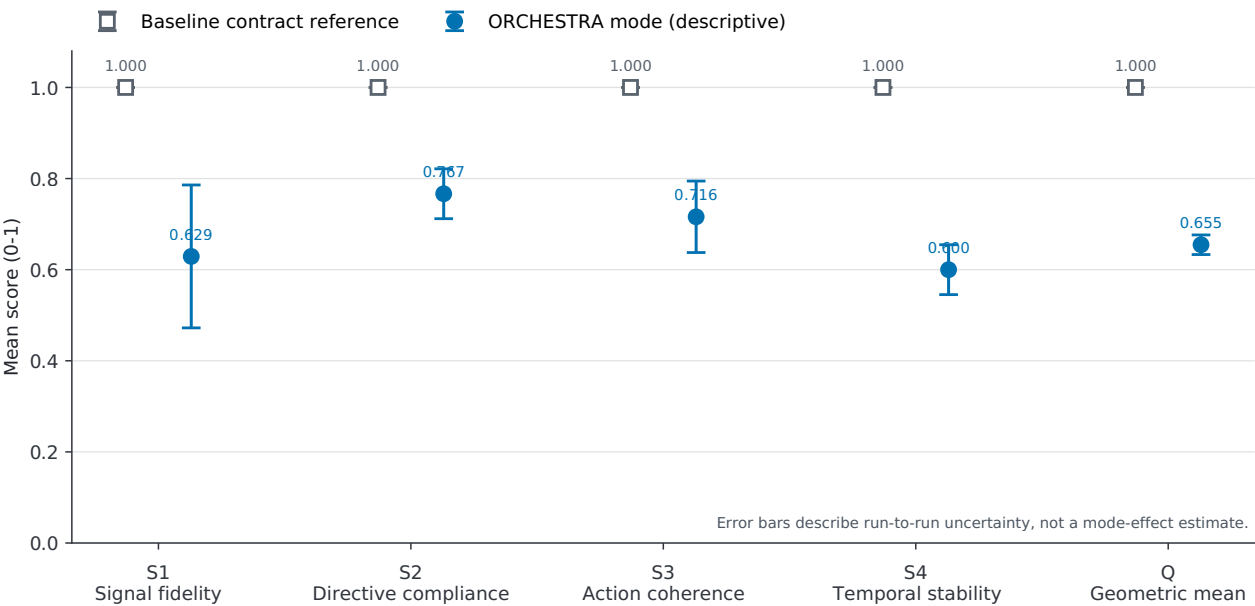
The intervals below are untruncated two-sided 95% Student-t intervals over the three invocation means.

Mode	S1 mean [95% CI]	S2 mean [95% CI]	S3 mean [95% CI]	S4 mean [95% CI]	Q mean [95% CI]
Observed-state reference	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
ORCHESTRA	0.629 [0.472, 0.786]	0.767 [0.712, 0.821]	0.716 [0.638, 0.794]	0.600 [0.545, 0.655]	0.655 [0.633, 0.676]

Visual summary

Coordination metrics across validated userspace runs

Run means with two-sided 95% t intervals (n=3 invocations per mode)

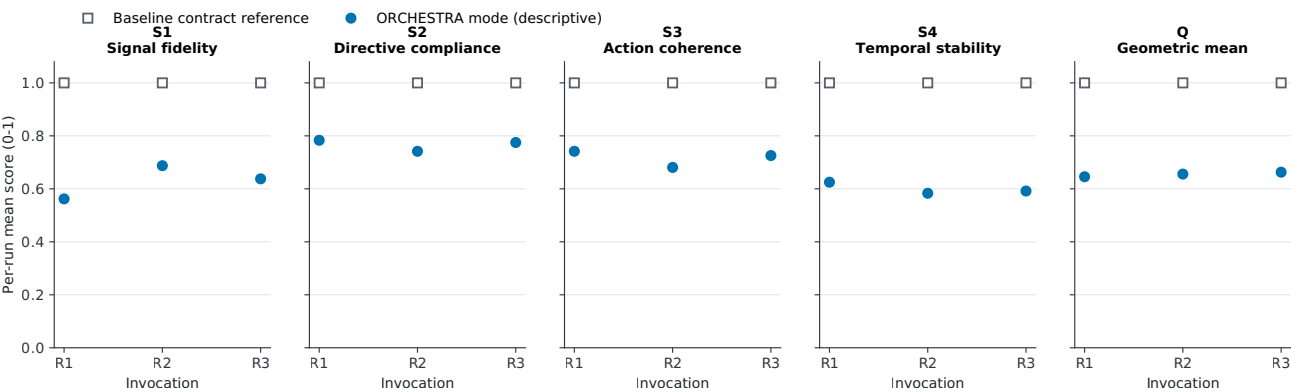


Claim boundary: descriptive userspace observations; baseline is a contract reference.
Modes are unpaired/endogenous - no causal, kernel-scheduler, or general-performance inference.

The reference values are contract checks and the modes are descriptive, unpaired, and endogenous. The intervals describe run-to-run uncertainty; they are not mode-effect intervals.

Run-level coordination metric variability

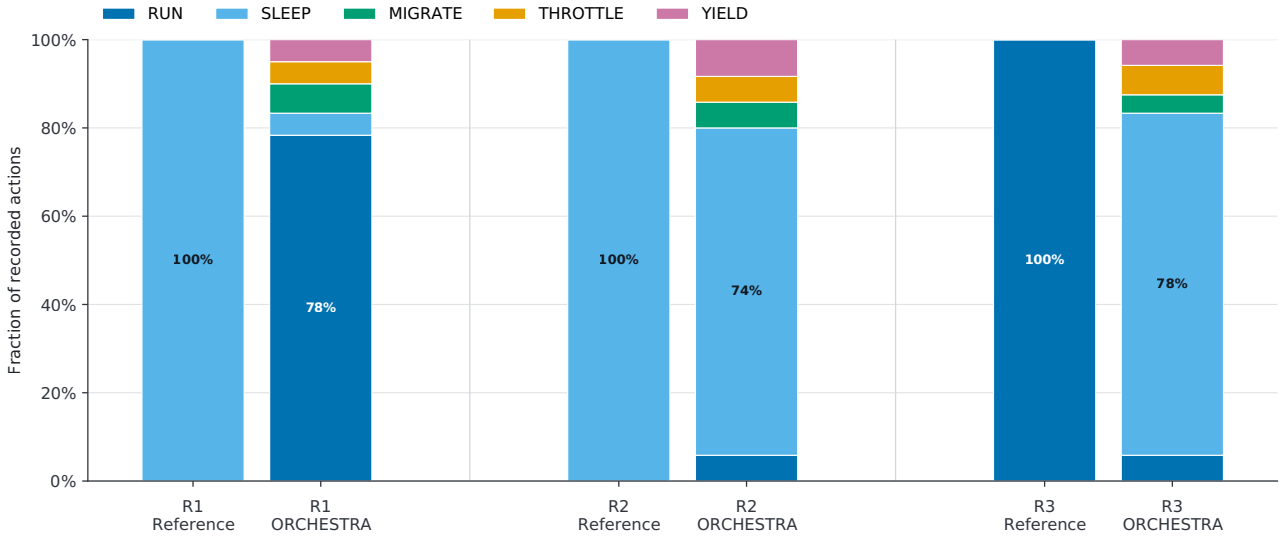
Each point is one validated invocation mean after the declared warm-up



Claim boundary: descriptive userspace observations; baseline is a contract reference.
Modes are unpaired/endogenous - no causal, kernel-scheduler, or general-performance inference.

Action selection mix by validated invocation

Fractions are calculated within each run after the declared warm-up



Claim boundary: descriptive userspace observations; baseline is a contract reference.
Modes are unpaired/endogenous - no causal, kernel-scheduler, or general-performance inference.

The plotting code, source-integrity hashes, deterministic-generation details, and figure-specific interpretation boundaries are recorded in the [figure manifest](#).

What the charts prioritize

- $S4 = 0.600$ is the lowest ORCHESTRA component mean. The matching bounded actuators are perceptual jitter and switching penalty, with migration bursts, latency, fairness, and responsiveness checked before accepting a change.
- $S1 = 0.629$ has the widest interval. That calls for independent held-out predictor calibration, confidence validation, and freshness/cost measurement; it does not justify retuning the predictor from aggregate Q .
- $S3 = 0.716$ supports evaluating bounded switching penalty and compatible policy consensus, while checking homogenization and correlated-failure risk.
- The action mix changes sharply across invocations. That reinforces the need for identical workload replay and counterbalanced order before comparing modes; it is not evidence of a treatment effect.

ORCHESTRA used the prediction for 95.6% of post-warm-up decisions on average; low-confidence rows used observed CPU. The reference mode deterministically selected the observed-state directive, which makes S2 perfect by construction. Its stable unanimous actions also make S3 and S4 perfect, while its observed signal reference normalizes S1. These values are a contract check, not evidence that the reference is a superior scheduler.

The host signal was endogenous: one reference invocation selected RUN for most post-warm-up actions and consumed much more userspace CPU than the other two reference invocations. This is direct evidence that subtracting the mode means would be confounded. No speedup, treatment effect, hypothesis test, or Linux scheduler improvement is reported.

Control decision

The implemented outer controller acts on the 20-tick schedule only when the current acceptance snapshot is complete, and controls only population mechanisms with a declared causal connection:

- deficient S4 increases perceptual jitter;
- deficient S3 or S4 increases the switching penalty;
- deficient S3 increases bounded Q-table consensus;
- healthy S3/S4 slowly decay the corresponding actuators; and
- actuator updates use $\text{beta} = 0.02 / \sqrt{1 + \text{step}}$ with fixed bounds.

The controller does not drive aggregate Q directly. It does not retune predictor gain when S3 or S4 is deficient. S1/prediction deficiency instead causes observed-state fallback; predictor or sampling changes require separate held-out calibration and budget evidence. Rejected or incomplete frames remain visible in coordination telemetry but cannot assign rewards or update the controller/consensus path.

Testing evidence and remaining gates

The associated test pass covered strict GCC/Clang warnings, GCC ASan/UBSan, 13 named C unit groups under both compilers, four benchmark-validator tests, five bounded integration scenarios, rejected-frame controller gating, tamper rejection, exact-zero Q, thundering-herd S4, reward dominance, signal teardown, and GCC/Clang static analysis.

The next performance experiment still needs an isolated and pinned host, counterbalanced or randomized mode order, an independently held-out calibration trace, workload replay that is identical across modes, a power-analysis-based repetition count, latency/fairness/energy measurements, and explicit instrumentation overhead. Kernel dispatch, real-time precedence, multi-core or NUMA hierarchy, key lifecycle, fault recovery, and long-duration reliability remain upstream work-package gates.

Those dependencies, controller guardrails, benchmark levels, stop criteria, rollback requirements, and the ordered next execution tranche are defined in the [release-readiness plan](#).

ORCHESTRA-OS Release-Readiness Plan

- Status: Proposed evidence-gated operations plan
- Date: 2026-08-03
- Scope: Research artifacts, kernel prototypes, experiments, scale-out work, and any future deployment-readiness review
- Governing source: repository `AGENTS.md`, including the WP1-WP10 dependency order and exit gates

1. Readiness decision today

The highest maturity class supported by current repository evidence is **Userspace-validated**. That classification applies only to the bounded, single-host userspace mechanisms that were actually exercised. It does not promote the whole architecture, or any untested mechanism, to that class.

Claim class	Current decision	Evidence boundary
Userspace-validated	Supported, with bounded scope	Real processes, real host observations, focused unit/integration tests, and the six-run <code>paper_cpu_smoke_v1</code> campaign exercise selected userspace WP2-WP6 precursor mechanics.
Kernel-prototyped	Not supported	There is no bootable test kernel, approved scheduling-extension implementation, kernel scheduler hook, Kconfig integration, or kernel selftest evidence.
Experimentally validated	Not supported	The current smoke is explicitly descriptive, unpaired, endogenous, and run on an active desktop with <code>n = 3</code> invocations per mode. It was not designed to estimate a causal treatment effect or scheduler improvement.
Deployment-ready	Not supported	WP1-WP10 exit gates, production key lifecycle, kernel recovery, controlled performance evidence, scale tests, threat validation, soak testing, and a formal readiness review have not been completed.

The current evidence base includes:

- the userspace state, reward, metric, and controller contract in [ADR 0001](#);
- the single-host canonical frame and verification contract in [ADR 0002](#);
- strict compiler, sanitizer, unit, and bounded integration checks reported by the current smoke result; and
- the versioned [smoke protocol](#) and [2026-08-03 result](#).

That result reports six successful short invocations and useful contract-level telemetry. It also records confounding, an active non-isolated host, no real-time workers, no kernel dispatch, and missing latency, fairness, energy, scale, fault-recovery, and long-duration evidence. A byte-identical local copy now exists under `artifacts/test-results/2026-08-03/` with an inventory and checksum manifest. Approved long-term storage, repository revision identity, retention, and retrieval policy remain unresolved. These limitations are release inputs, not footnotes to waive.

There is also a canonical-scope hygiene blocker. The legacy `orchestra_real_cpu_demo/` implements process groups, group runnable budgets, and coordinator-driven restart behavior that the approved research scope explicitly excludes from the canonical architecture. Before even a U0 research snapshot is packaged, this artifact must either be moved under `research/experiments/` with an **Exploratory** claim, explicit hypothesis, and separate documentation, or be excluded from the release manifest. It must not be built, measured, or described as part of canonical ORCHESTRA-OS.

Evidence hygiene also applies to the browser demo. `orchestra_os_demo.html` describes rotating-key HMAC behavior, but its integrity routine is an illustrative random tamper/block model rather than a cryptographic implementation. It may remain only as clearly labeled simulation/illustration and must not be cited as authentication, key-rotation, tamper-detection, or production-security evidence. A release inventory must classify or exclude every top-level executable/demo; the current root Makefile tests only the paper CPU demo and is not a complete release inventory.

2. Meaning of a release

A release is an immutable, reproducible research artifact with an explicit claim class. A version number, passing build, demo, or short benchmark does not by itself change maturity. Each release record must state:

1. exactly which artifact and commit or content hashes were evaluated;
2. the strongest permitted claim class and its scope;
3. every applicable gate as passed, failed, not run, or not applicable;
4. failures, exclusions, exceptions, and approved risk decisions;
5. links or checksums for manifests, raw data, processed data, and reports;
6. the environment and configuration needed to reproduce the result; and
7. the tested rollback path and the evidence that it restored safe operation.

Research snapshots may be released while later gates remain open, but their names and release notes must say so. A research snapshot must not be presented as a production candidate.

3. Dependency and status map

No downstream claim may bypass its upstream evidence gate. Work may proceed in parallel, but a later work package remains blocked for release purposes until its dependencies pass.

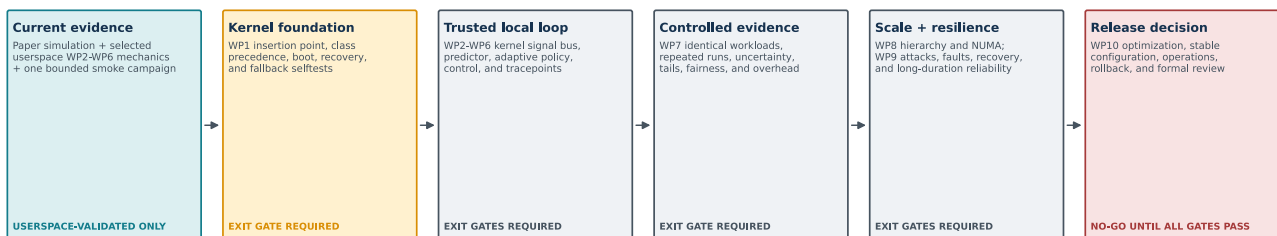
```
flowchart TD
    U["Current evidence<br/>Userspace-validated only"]:::current
    W1["WP1 kernel foundation<br/>gate not passed"]:::closed
    W2["WP2 signal bus<br/>gate not passed"]:::closed
    W3["WP3 predictor<br/>gate not passed"]:::closed
    W4["WP4 adaptive scheduling<br/>gate not passed"]:::closed
    W5["WP5 measurement/control<br/>gate not passed"]:::closed
    W6["WP6 instrumentation<br/>gate not passed"]:::closed
    W7["WP7 evaluation<br/>gate not passed"]:::closed
    W8["WP8 scale-out<br/>gate not passed"]:::closed
    W9["WP9 security/reliability<br/>gate not passed"]:::closed
    W10["WP10 readiness review<br/>gate not passed"]:::closed

    U -. "precursor evidence" .-> W1
    W1 --> W2 --> W3 --> W4 --> W5 --> W6 --> W7 --> W8 --> W9 --> W10

    classDef current fill:#d9f2d9,stroke:#237a23,color:#111;
    classDef closed fill:#f4dddd,stroke:#9c2f2f,color:#111;
```

ORCHESTRA-OS: evidence-gated path to a release decision

Stages are dependency gates, not elapsed time or percent complete. A version label cannot replace evidence.



Release meaning: Deployment-ready only after correctness, safety, security, performance, reliability, reproducibility, observability, and operational gates pass.

Current smoke results guide engineering priorities; they do not establish kernel scheduling benefit, causal performance, hard-real-time behavior, or production security.

The boxes are dependency gates, not elapsed time or percent complete.

4. Proposed release stages

Stage names describe review checkpoints, not schedules. No calendar estimate is implied.

Stage	Work-package scope	Go evidence	Permitted claim after approval	Required rollback before approval
U0 - bounded userspace research snapshot	Existing simulator/ userspace precursors	Reproducible build and tests; immutable smoke inputs and retained raw artifacts; limitations disclosed	Userspace-validated only for named, tested mechanisms and environments	Bounded process termination, child/shared-resource cleanup, and artifact preservation
K0 - kernel foundation candidate	WP1	Bootable test kernel or approved extensible-scheduler implementation; conventional and real-time fallback regression tests; documented interfaces; automated build, boot, and recovery	Kernel-prototyped only for the WP1 mechanisms actually exercised	Known-good boot entry, runtime disable/detach where supported, and tested conventional-scheduler recovery
K1 - trusted local scheduling substrate	WP2-WP3 after WP1	Coherent kernel signal publication, rejection and recovery tests, key lifecycle design, measured verification budget, held-out predictor calibration, confidence fallback, and stale-prediction exclusion	Kernel-prototyped local signal/predictor path; no adaptive-performance or production-security claim	Disable signal/prediction path; reject incompatible schemas; return to observed-state and then conventional scheduling
K2 - bounded adaptive kernel prototype	WP4-WP6 after WP1-WP3	Exact kernel action contracts, Hybrid Safety Layer, real-time/exempt bypass, bounded authority, S1-S4/controller tests, saturation response, selective tracing, and measured instrumentation overhead	Kernel-prototyped adaptive path under declared test conditions	Per-task removal, global adaptive disable, policy reset, safe scheduler fallback, and known-good boot
E0 - experimental validation candidate	WP7 after WP1-WP6	Predeclared controlled experiments, identical workload/hardware conditions for baselines, sufficient repetitions, uncertainty, complete failure reporting, and traceable raw-to-report lineage	Experimentally validated only for the preregistered questions, workloads, hardware, and confidence bounds that pass	Preserve previous candidate; stop on validity breach; rerun only under a new immutable manifest
S0 - scale and assurance candidate	WP8-WP9 after WP7	Multi-level failure isolation, measured communication/ NUMA costs, documented consistency, current threat model, attack/fault campaigns, recovery distributions, and long-duration reliability evidence	Experimentally validated at the tested scale and assurance envelope; still not deployment-ready	Higher-tier isolation with safe local continuity, key revocation/recovery, fault containment, and tested downgrade
R0 - deployment-readiness candidate	WP10 after WP9	Stable defaults, sustained-run criteria, bounded optimization, operational documentation, complete configuration/rollback evidence, no unresolved critical vulnerability without explicit risk decision, and formal review	Deployment-ready only after the formal review grants that class and records its exact envelope	Operational rollback rehearsal, compatible data/ schema handling, known-good software/kernel image, and post-rollback verification

If a stage changes a public protocol, scheduler precedence, action semantics, state schema, reward, compliance rule, metric, controller, fallback, predictor, or key lifecycle, an approved ADR and migration/reset plan are entry requirements.

5. WP1-WP10 evidence ledger

“Precursor evidence” means useful userspace research input. It does not satisfy a kernel or downstream exit gate.

WP	Dependency	Evidence present now	Evidence still required to pass the exit gate	Release status
WP1 - Linux Kernel Integration	Simulation/userspace evidence	Userspace action approximations and Linux-host measurements	Approved insertion point and precedence ADR; kernel or approved extension implementation; task/run-queue interfaces; boot/build/debug/recovery automation; conventional, deadline, and real-time fallback regressions; kernel smoke tests	Not started for gate purposes; not passed
WP2 - Signal Bus	WP1	Userspace 128-byte payload, HMAC verification, coherent single-writer snapshots, freshness/sequence checks, tamper tests, and bounded last-known-good behavior; schema 1 lacks the required prediction-horizon field	Kernel-safe fixed-point/frame design with prediction horizon; core/socket/NUMA/system hierarchy; complete concurrency and lifetime proof; explicit duplicate rejection/diagnostics; key provisioning/rotation/revocation/recovery; live epoch-transition and rotation-race tests; parser fuzzing; contention and recovery tests; publication/verification mean and tail budget	Userspace precursor only; not passed
WP3 - Predictive Engine	WP2	Lightweight userspace estimator, confidence field, observed-state fallback, and forecast telemetry	Independent held-out calibration/evaluation; versioned parameter artifact; workload/hardware metadata; no-look-ahead proof; recalibration triggers; degraded/negative cases; mean/tail latency, CPU, memory, and cache budget in the intended scheduler path	Userspace precursor only; not passed
WP4 - Adaptive Scheduling	WP3	State schema v2, canonical five-action selection, bounded reward terms, annealed exploration, consensus precursor, and userspace real-process execution	Kernel contracts for RUN/SLEEP/MIGRATE/THROTTLE/YIELD; admission and removal; safety-class precedence; syscall/dispatch outcome evidence; starvation/fairness controls; conventional fallback; bounded safe exploration; stress evidence without uncontrolled oscillation	Userspace precursor only; not passed
WP5 - Measurement and Control	WP4	Bounded S1-S4 and zero-preserving geometric-mean Q; thundering-herd test; 20-tick, decaying, causally mapped userspace controller	Per-level metrics; versioned compliance rule; longer-window/stability analysis; sustained-saturation alert; controller disable, rollback, and re-entry; convergence and oscillation tests; kernel actuator effects; false-good tests across integrated behavior	Userspace precursor only; not passed
WP6 - Instrumentation	WP5, with instrumentation developed alongside prior packages	Machine-readable userspace CSV for predictions, actions, S1-S4, Q, controller state, rejections, fallbacks, and missed deadlines	Selective kernel tracepoints for final dispatch and all actions; reason-specific verifier/fallback/controller events; coherent timestamps; schema metadata; dropped-event accounting; mean/tail tracing overhead and perturbation evidence	Partial userspace observability only; not passed

WP	Dependency	Evidence present now	Evidence still required to pass the exit gate	Release status
WP7 - Experimental Evaluation	WP6 and stable WP1-WP5 semantics	Versioned bounded smoke manifest, schema validator, three runs per mode, uncertainty summaries, hashes, and disclosed failures/limitations	Isolated and pinned host; counterbalanced/randomized order; identical replayable workload across baselines; independent calibration data; predeclared primary outcomes and power/repetition rationale; latency, fairness, energy, throughput, overhead, and robustness; Linux baseline under identical conditions; confirmatory analysis	Smoke evidence only; not passed
WP8 - Scalability and Multi-Level Coordination	WP7	No qualifying implementation; only a fixed core-local tier identity	Explicit core/socket/NUMA/node/cluster responsibilities and conflicts; distributed ordering/clock/consistency design; local continuity under higher-tier loss; scale tests; communication and remote-memory costs; bottleneck and limit report	Not started for gate purposes; not passed
WP9 - Security, Reliability, and Resilience	WP8	Focused userspace HMAC/tamper/freshness tests and bounded fallback precursor	Current threat model and trust boundaries; production key lifecycle; spoof/replay/corruption/authorization campaigns; combined attack/fault injection; recovery-time distributions; long-duration leak/instability tests; security monitoring; explicit critical-risk decisions	Focused precursor tests only; not passed
WP10 - Optimization and Deployment Readiness	WP9	No qualifying integrated evidence	Profile-led optimization without semantic/security regression; stable defaults; sustained-run acceptance; configuration and rollback documentation; operator diagnostics; end-to-end regression; explicit limitations; formal readiness decision	Blocked by WP1-WP9; not passed

6. What the controller may control

Control changes must follow diagnosed submetrics and causal evidence. Aggregate **Q** is an outcome and alarm signal; it is not sufficient diagnosis for choosing an actuator.

Observed deficit	Permitted control path	Required guard/evidence	Disallowed shortcut
S1 signal fidelity/freshness	Reject invalid/stale data; use bounded last-known-good data; fall back to observed state; adjust prediction horizon/gain only when forecast quality is the diagnosed cause; adjust sampling only when freshness/overhead analysis supports it	Confidence, age, verification result, prediction error after observation, deadline and cost telemetry	Retuning prediction because S3/S4 or aggregate Q is low
S2 directive compliance	First diagnose eligibility, exemptions, directive/reward/state consistency, exploration, action failure, and final dispatch; any learning-authority change requires a bounded, versioned policy experiment	Versioned compliance rule; selected-versus-effective action; reward components; safety overrides; fairness and responsiveness	Raising agreement by overriding safety classes, hiding exemptions/failures, or making exploration unbounded

Observed deficit	Permitted control path	Required guard/evidence	Disallowed shortcut
S3 action coherence	Bounded, infrequent compatible-policy consensus and a bounded switching penalty	Schema compatibility, participant/timeout telemetry, homogenization and correlated-failure tests, performance/fairness checks	Treating uniform action as inherently good or performing exact counterfactual work in a kernel hot path
S4 temporal stability	Perceptual jitter derived from robust observation-noise estimates and a bounded switching penalty	Switching bursts, migration bursts, latency, fairness, responsiveness, and run-to-run variance; controller rate slower than learner	Maximizing stability by making processes ignore legitimate signal changes
Latency, fairness, energy, or overhead regression	Diagnose the responsible scheduler/action/instrumentation path and use its specifically approved bounded control	Separate service-level telemetry and predeclared limits	Optimizing Q as a proxy for these independent outcomes

Every automatic actuator must have a minimum, maximum, rate limit, update cadence, saturation counter, alert, rollback/disable condition, re-entry rule, and telemetry for requested and applied values. Outer-loop updates must remain slower than inner learning. Reward logic, state boundaries, directive logic, compliance scoring, tests, documentation, and experiment version must change together.

The current controller is a userspace precursor: it updates every 20 ticks with a decaying step and fixed bounds, but it consumes an instantaneous eligible snapshot and lacks sustained-saturation counters, alerting, rollback, and disable/re-entry state. Those are explicit K2/WP5 blockers.

7. Benchmark evidence ladder

Benchmarking advances a claim only when the design can answer the declared question. Thresholds must be defined in an approved protocol or manifest before data collection; they must not be selected after seeing results.

Level	Question	Minimum controls and outputs	Stop/go interpretation
B0 - contract smoke	Does the artifact run and preserve local invariants?	Fixed inputs, schema validation, bounded duration, raw output, errors, hashes, sanitizer/unit/integration tests	Current campaign supports this level only. Failures stop the artifact release; success does not imply performance improvement.
B1 - controlled userspace characterization	What costs and behaviors arise from the userspace mechanisms on declared hardware/workloads?	Isolated/pinned host; fixed frequency policy; randomized or counterbalanced order; identical workload replay; held-out calibration; repetitions justified before the run; latency, throughput, fairness, CPU, memory, energy, prediction, S1-S4/Q, fallback, and instrumentation overhead	Go only for scoped userspace characterization. Do not infer kernel scheduling effects.
B2 - kernel functional and safety validation	Do kernel semantics, precedence, and fallbacks match their contracts?	Kernel selftests and stress tests; all five effective action outcomes; deadline/RT/exempt bypass; starvation checks; invalid/stale signal cases; boot and runtime rollback; crash/hang capture	Any precedence, safety, liveness, or recovery failure stops kernel-candidate promotion.
B3 - kernel comparative evaluation	Does the integrated prototype improve predeclared outcomes without unacceptable regressions?	Same kernel base, hardware state, workload traces, affinity, warm-up, and instrumentation across Linux baselines and ORCHESTRA; randomized order; sufficient independent runs; tail distributions and uncertainty; failed-run accounting	Promote only the preregistered claims that meet their predeclared criteria. A Q improvement cannot compensate for a breached safety, tail-latency, fairness, or overhead limit.
B4 - scale, fault, and adversarial validation	Where does coordination remain safe and useful under topology, delay, loss, fault, and attack?	Multi-core/NUMA/node scale matrix; communication and remote-memory costs; clock/order cases; partitions; tamper/replay/spoof/key events; combined faults; recovery distributions	Any unsafe loss of local continuity, trust bypass, or unbounded controller behavior stops scale/assurance promotion.

Level	Question	Minimum controls and outputs	Stop/go interpretation
B5 - sustained readiness validation	Can the approved configuration operate, degrade, recover, and roll back for the declared service envelope?	Long-duration workload mix; memory/resource leak checks; repeated failure/recovery; operational alarms; upgrade/downgrade; rollback rehearsal; full provenance	Evidence feeds WP10 formal review. It cannot self-grant Deployment-ready status.

For every benchmark, the invocation - not a tick or worker row - is the default independent statistical unit unless the protocol justifies another model. Report warm-up, missing data, exclusions, timeouts, crashes, unsuccessful privilege changes, and all failed runs. Keep raw data immutable and perform validation and processing in separate, versioned steps.

8. Universal go and stop criteria

Go criteria

A candidate may enter its next stage only when all of the following are true:

- every upstream exit gate is recorded as passed against immutable evidence;
- the candidate's interfaces and semantics are frozen or changed through an approved ADR with compatibility and migration handling;
- functionality, safety, security, reliability, observability, performance, and reproducibility evidence covers the declared envelope;
- all predeclared tests ran, and failures and exclusions are included in the decision record;
- rollback was exercised from the exact candidate configuration;
- raw artifacts and their checksums are retained under the artifact policy;
- open limitations are consistent with the proposed claim class; and
- the applicable review explicitly approves the scoped promotion.

Immediate stop criteria

Stop promotion, preserve evidence, and diagnose before rerunning when any of the following occurs:

- a proposed claim is stronger than its evidence class;
- a release inventory omits an executable, demo, test surface, generated artifact, or explicitly excluded exploratory component;
- an upstream gate is missing, failed, or has unreviewed incompatible changes;
- the artifact, manifest, schema, environment, or raw-data identity cannot be established;
- raw data were overwritten, silently filtered, or cannot be traced to the processed result;
- a hard-real-time, deadline, exempt, or conventional task bypasses its required precedence or fallback contract;
- a torn, stale, replayed, malformed, wrong-source, or unauthenticated frame is accepted for a new adaptive decision;
- a stale or low-confidence prediction affects a decision outside its approved fallback policy;
- S1-S4 or Q becomes non-finite/out of range, exact-zero collapse is lost, or a thundering herd receives a false-good stability score;
- an actuator exceeds its bounds/rate, oscillates without containment, remains saturated without an alert, or lacks a tested rollback;
- a kernel panic, hang, data corruption, starvation event, uncontrolled migration/throttle burst, or unrecoverable scheduler failure occurs;
- an unresolved critical vulnerability lacks an explicit recorded risk decision;
- instrumentation loss or perturbation makes the declared conclusion invalid; or
- a predeclared safety, tail-latency, fairness, overhead, recovery, or stability limit is breached.

A stop is an experimental outcome. It must remain in the run ledger and must not be converted into an exclusion merely to improve the reported result.

9. Security and reliability requirements

Before WP9 can pass, the program must have an approved threat model covering producer, consumer, key holder, control plane, instrumentation, storage, and core/node/cluster trust boundaries. The inherited userspace master key is test scaffolding, not a production root of trust.

Required evidence includes:

- generation, provisioning, per-tier derivation, rotation overlap, revocation, node join/leave, compromise recovery, audit, and secret-erasure behavior;
- bounds/schema/identity/sequence/freshness/authentication verification in the required order, constant-time tag comparison, and parser/verifier fuzzing;
- tamper, replay, duplicate, stale, wrong-source, unauthorized modification, sequence/epoch boundary, partial-publication, and key-transition tests;
- communication, publisher, reader, predictor, policy, controller, storage, and clock fault injection, including combined attack-plus-fault cases;
- safe local continuity when prediction, signal service, controller, or a higher hierarchy tier fails;
- mean and tail detection/recovery time, availability impact, data loss, and re-entry behavior; and
- long-duration evidence for leaks, counter/epoch handling, progressive instability, livelock, starvation, and unbounded adaptation.

Detected integrity failures must fail closed for the adaptive input while scheduling fails operationally safe: bounded last-known-good use, observed-state fallback where valid, conventional Linux scheduling when coordination is unsafe, and explicit recovery/re-entry conditions.

10. Rollback and recovery requirements

Rollback is part of the feature, not an operator assumption.

- **Userspace:** terminate within a bounded watchdog window; reap children; release mappings, descriptors, affinity changes, and other resources; retain stdout/stderr and partial artifacts with a failure record.
- **Kernel:** provide a known-good boot selection and documented recovery console; support a tested build-time exclusion and, where the implementation permits it, a runtime global disable or scheduling-extension detach; preserve conventional and real-time scheduling precedence during failure.
- **Task/policy:** remove eligibility safely, stop new adaptive decisions, complete or cancel in-flight transitions coherently, and reset or migrate versioned policy state without interpreting incompatible tables.
- **Signal/predictor/controller:** reject incompatible schemas, expire unsafe data, disable the affected component, bound last-known-good use, and verify observed-state/conventional fallback before re-entry.
- **Hierarchy:** isolate a failed or untrusted higher tier and maintain safe node/core-local scheduling; reconnect only under explicit identity, ordering, freshness, and state-reconciliation rules.
- **Release:** preserve the previous signed/hashed artifact and configuration; exercise upgrade, downgrade, and data/schema compatibility; verify health and scheduler policy after rollback.

Rollback tests must cover failure during activation, steady operation, schema or key transition, and recovery. A procedure that has only been documented but not successfully exercised does not satisfy a go gate.

11. Observability requirements

The existing CSV is useful userspace evidence, but release candidates require selective, versioned instrumentation that records what the system actually did, not only what a policy selected. The applicable stage must expose:

- observation, prediction, confidence, age, horizon, error when knowable, and observed-state fallback;
- frame schema, tier/source, sequence/epoch, acceptance/rejection reason, coherent-read retry/exhaustion, verification latency, and last-known-good age;
- eligibility/exemption, requested action, safety override, effective kernel action/dispatch outcome, migration target/result, throttle duration, sleep, yield, wake, preemption, and action failures;
- per-process/core/node/system S1-S4 and Q with metric/compliance schema versions;
- controller input window, skip reason, step/cadence, requested/applied actuator, bounds, rate limiting, saturation streak, alert, rollback, and re-entry;

- fairness, starvation, run-queue, latency, throughput, CPU, memory, cache, energy/thermal, network/I/O where relevant, and deadline-miss measures; and
- failure, recovery, dropped-event, clock-domain, environment, and artifact provenance fields.

Trace collection must be selectively enabled. Timestamp comparability and clock scope must be declared, dropped records must be visible, output must carry its schema and metadata, and tracing's mean/tail CPU, memory, cache, and latency perturbation must be measured. If instrumentation changes the behavior beyond a predeclared bound, the affected benchmark cannot support a go decision.

12. Reproducibility and artifact requirements

Every release-qualifying campaign must retain:

- repository revision and dirty state, or content hashes when Git metadata is unavailable;
- compiler/toolchain, dependencies, build command, flags, kernel source/config, firmware/microcode, and binary/module hashes;
- hardware topology, NUMA layout, memory, storage, network, clock source, virtualization/container state, power/frequency/turbo/thermal policy, privilege/capabilities, affinity, background-load controls, and time zone;
- immutable experiment manifest, workload/input hashes, seeds, calibration artifact and split, schema versions, baseline order, warm-up, timeout, and repetition rationale;
- per-invocation status, stdout/stderr, raw traces, validation results, exclusion reasons, processed outputs, analysis configuration/code, and report; and
- checksums and lineage linking every report value to raw evidence.

Raw data must never be mutated. Large artifacts may live outside the repository under the approved artifact policy, but the repository must retain durable identifiers, checksums, schema, retrieval instructions, and retention status. Temporary paths alone are insufficient for a release gate. A rerun after code, manifest, environment, or analysis changes is a new campaign, not a replacement for the earlier record.

13. Formal review record

The review record for each stage must include:

1. proposed stage and exact claim envelope;
2. WP gate checklist and links/checksums for each item of evidence;
3. benchmark questions, predeclared criteria, results, uncertainty, failures, exclusions, and negative findings;
4. security, reliability, observability, and rollback assessments;
5. open defects and limitations, with explicit risk decisions where permitted;
6. compatibility, policy reset/migration, and operational recovery decisions;
7. reviewer roles and the approve, reject, or return-for-evidence decision; and
8. the stronger claims that remain prohibited after the decision.

Only the formal WP10 review may grant **Deployment-ready**, and only for the configuration and operating envelope named in that record.

14. Next execution tranche

This is the shortest evidence-preserving route to the first kernel candidate. Items are ordered by dependency, not duration.

Order	Deliverable	Completion evidence	Dependency
1	Freeze the release surface and claim ledger	Root release inventory classifies every source, binary, historical artifact, and demo; the noncanonical group demo is isolated/excluded; the browser demo is labeled non-cryptographic; root README, license, security, contribution, artifact-retention, and support policies exist	None

Order	Deliverable	Completion evidence	Dependency
2	Establish immutable provenance	Version-control identity, reproducible toolchain/build recipe, clean release manifest, source-to-binary attestation, and approved long-term storage/retrieval policy building on the retained local checksums	Order 1
3	Approve the WP1 kernel design	ADR selects native versus extensible scheduler insertion, class precedence, task/run-queue state, locking/lifetime rules, action/fallback contracts, debug strategy, and known-good boot recovery	Orders 1-2; simulation/userspace evidence
4	Build the recoverable K0 foundation	Automated kernel/extension build, disposable boot target, attach/detach or disable path where applicable, conventional/deadline/RT regression suite, kernel smoke tests, and exercised rollback	Order 3
5	Approve the WP2 trust and frame revision	Threat model plus schema ADR adds prediction horizon and explicit duplicate/replay/wrap behavior; defines key ownership/lifecycle, coherent-read retry diagnostics, hierarchy interfaces, and hot-path budgets	Order 4; design work may start earlier but cannot pass first
6	Implement and stress the local kernel signal substrate	Parser/verifier fuzzing, high-contention publication/read tests, epoch transition and rotation-race tests, publisher-loss recovery, no torn accepted frames, and measured p50/p95/p99/max cost	Order 5
7	Freeze the WP3 evaluation split and model artifact	Immutable held-out traces, versioned parameters, confidence calibration, degraded/OOD cases, negative-Kalman regression, and scheduler-budget measurements	Order 6
8	Enter K1 review	WP1-WP3 ledgers complete, fallbacks and rollback exercised, all failures retained, and formal review approves only the tested kernel signal/predictor envelope	Orders 1-7

WP4-WP10 implementation follows only after these upstream contracts are stable. Performance tuning may be prototyped for cost discovery, but no optimization result may bypass semantic, safety, or validity gates.

15. Open uncertainties that block stronger claims

The roadmap deliberately does not guess answers to these unresolved decisions:

- the approved Linux scheduler insertion point, class precedence, and whether an extensible scheduling framework or a native class will satisfy WP1;
- kernel-safe frame encoding, synchronization/lifetime model, hierarchy layout, and publication/verification budgets;
- root of trust, provisioning authority, node identity, rotation overlap, revocation, and compromise-recovery model;
- independent calibration dataset, target workload families, prediction budget, confidence calibration, and recalibration trigger;
- exact kernel semantics and safety limits for all five canonical actions;
- the versioned S2 compliance rule when safety overrides or action failures occur, and the approved approximation for difference rewards in kernel paths;
- controller input-window design, saturation duration, alarm thresholds, rollback/re-entry policy, and quantitative stability criteria;
- primary experimental outcomes, smallest relevant effects, power/repetition rationale, baseline set, and acceptance limits for latency, fairness, energy, overhead, recovery, and stability;

- target hardware/topology and the distributed clock, consistency, conflict, and partition model for WP8;
- approved long-term artifact storage, retention, access, and retrieval policy; the current smoke now has a checksummed local copy but no approved durable external store or repository revision identity; and
- deployment environment, support model, sustained-run duration, and formal review authority.

Resolve each architecture-level uncertainty through the required ADR or versioned experiment protocol before using it as a release criterion. This plan contains no calendar estimates because the evidence gates, not elapsed time, determine readiness.

ORCHESTRA-OS test evidence - 2026-08-03

This directory contains every text/data test artifact that remained available locally when the archive was created, plus a fresh full regression capture.

Evidence map

Path	Classification	Interpretation
benchmark/authoritative-c/	Authoritative userspace smoke	Exact copy of <code>/tmp/orchestra-paper-cpu-smoke-v1-20260803-c</code> ; 6/6 validated invocations, source hash <code>d5ec68...8713</code> ; this is the campaign used by the result report and figures.
benchmark/superseded-a/	Superseded	Exact copy of complete six-run campaign <code>-a</code> , source hash <code>b03b78...c55a</code> ; it predates final implementation fixes and must not be mixed with the authoritative result.
benchmark/superseded-b/	Superseded	Exact copy of complete six-run campaign <code>-b</code> , source hash <code>5de386...2193</code> ; it predates final implementation fixes and must not be mixed with the authoritative result.
development-smoke/	Historical development evidence	CSV/log/stdout/stderr files copied from <code>/tmp/orchestra-os-testing.1InLRQ</code> ; provenance is incomplete, so these are debugging records rather than experiment evidence.
regression/final/	Current regression evidence	Fresh <code>make test</code> capture after artifact preservation support was added; includes the combined transcript, machine-readable result, and all five integration scenarios' CSV/log files.
regression/pre-repository-artifact-defaults/	Superseded regression capture	Passing run immediately before the plotting/reporting defaults were changed from <code>/tmp</code> to the retained repository artifact. Retained for completeness.
regression/pre-artifact-index-check/	Superseded regression capture	Passing run with preserved integration CSV/log files immediately before the artifact indexer was added to the top-level syntax checks. Retained for completeness.
regression/pre-integration-artifact-capture/	Superseded regression transcript	Passing run immediately before integration CSV/log preservation was added. Retained so no test result is silently discarded.
regression/pre-capture-tool-check/	Superseded regression transcript	Passing run immediately before the capture tool was added to the top-level syntax checks. Retained for completeness.

Current regression result

`regression/final/test-result.json` records return code `0` , unchanged input hashes, the compiler/tool versions, and the merged transcript hash. The run passed:

- 13/13 named GCC ASan/UBSan unit groups;
- the same 13/13 groups under Clang policy warnings;
- 4/4 benchmark-validator tests; and
- 5/5 integration scenarios covering baseline, ORCHESTRA cadence, tamper rejection, rejected-frame controller gating, and bounded signal teardown.

The regression suite is correctness evidence for the named userspace tests. It does not strengthen the system beyond the **Userspace-validated** claim class.

Copy and hygiene record

- All three benchmark directory copies passed recursive byte comparison against their `/tmp` sources.
- All 22 retained development CSV/log/stdout/stderr files passed byte comparison against their source files.
- Seven generated development test executables were deliberately not copied; they are rebuildable inputs, not test results. Their source hashes are listed below so the exclusion is auditable.
- A text scan found no cookie, authorization, bearer-token, password, API-key, client-secret, access-token, refresh-token, or email-address matches.
- Raw provenance retains the local hostname, username/path, process snapshot, and hardware metadata because this is a local research archive. Review and sanitize a derived copy before public distribution; do not mutate the raw evidence.
- The original `/tmp` directories were not deleted.

Excluded development executables:

File	SHA-256
<code>orchestra_atomic_clang</code>	<code>08ccc1796716e02a3dea367fd866a45209bd856fc75a6d85507bbcb7be2a5ad6</code>
<code>orchestra_atomic_gcc</code>	<code>a051f65088d5f07364967aad6c9936a1cedd2cee347a6853a7f4f4b3fced3c3d</code>
<code>orchestra_release</code>	<code>36520d79649dc240a805fe646bd325d430f2b884eca40479a1ab92897ecd8692</code>
<code>orchestra_san</code>	<code>03e23ad0b6df52f05eaa956622a5fde673efc082a20fd4f0e3a6064c52448f9</code>
<code>orchestra_v2</code>	<code>0aeca6bf63e9a01a6e522c5364306f7aa9f7c72eb8070e89c58c463b8a29884d</code>
<code>orchestra_v2_clang</code>	<code>74fd75d32793fa8b7c097eaa8b8d10eff8cea90473635bf28386c5283d5de7ee</code>
<code>orchestra_v2_strict</code>	<code>3cf28905af15f7455cd8ca8cfa07a004c629a5055690fcbe7e9b1917284f0f01</code>

`SHA256SUMS` covers every retained evidence file except the checksum and inventory files themselves. `inventory.json` records counts, sizes, category totals, and the checksum-manifest identity.